sdm-spatial-coverage

Exploratory analysis of observation coverage over SDM spatial domains

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1 What this does

When building a species distribution model (SDM), you choose a **spatial domain**, the area over which the model will predict. This domain is discretised into spatial units:

- **Triangles** if you use a mesh (e.g. sdmTMB or R-INLA)
- **Cells** if you use a regular prediction grid (e.g. GAM, GLM, or BART)

Before fitting an SDM, it is useful to check whether observations are well distributed across the prediction domain.

Script	Spatial unit	Core function	Use case
`mesh_coverage.R`	INLA mesh triangles	`run_mesh_coverage()`	sdmTMB, R-INLA
`grid_coverage.R`	Regular raster cells	`run_grid_coverage()`	GAM, GLM, BART

2 Why it matters

2.1 Too many observations per unit → nugget risk

If many observations fall within the same triangle or cell, the model may not resolve fine-scale variability well. In geostatistics, this is related to the **nugget effect**: variance at very short distances appears as noise because the model discretisation is too coarse relative to the data.

> Note
>
> This idea is closely related to the SPDE approach described by Lindgren et al. (2011), where the relationship between spatial resolution and short-scale variability is discussed in the context of Gaussian random fields and mesh-based approximations.
>
> **Reference** Lindgren, F., Rue, H., & Lindström, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential equation approach. *Journal of the Royal Statistical Society: Series B*, 73(4), 423–498.

2.2 Too few occupied units → extrapolation risk

If only a small fraction of units contains any observation, the model predicts mostly by extrapolation. This is not necessarily wrong, but it is useful to be aware of it before fitting the model.

3 Coverage metrics

For each spatial unit (triangle or cell), three summary metrics are computed:

Metric	Definition	What it indicates
% Occupied	Fraction of units with ≥ 1 observation	Geographic coverage of the prediction domain
Ratio	Total observations / occupied units	Mean data density per occupied unit
P95	95th percentile of observations per unit	Clustering — robust to extreme values

3.1 Practical interpretation

> ⚠ Warning
>
> These are simple heuristic ranges for exploratory interpretation only. They are not formal thresholds and should not be used as decision rules.

Metric	Range	Interpretation
Ratio	< 1.5	Too coarse — loss of spatial detail
Ratio	1.5–3	Good balance
Ratio	> 3	Too fine — many empty units
P95	< 5	Well distributed
P95	5–8	Moderate clustering
P95	> 8	High clustering
% Occupied	< 10%	Most of the domain unobserved
% Occupied	10–30%	Moderate geographic coverage
% Occupied	> 30%	Good geographic coverage

4 Examples

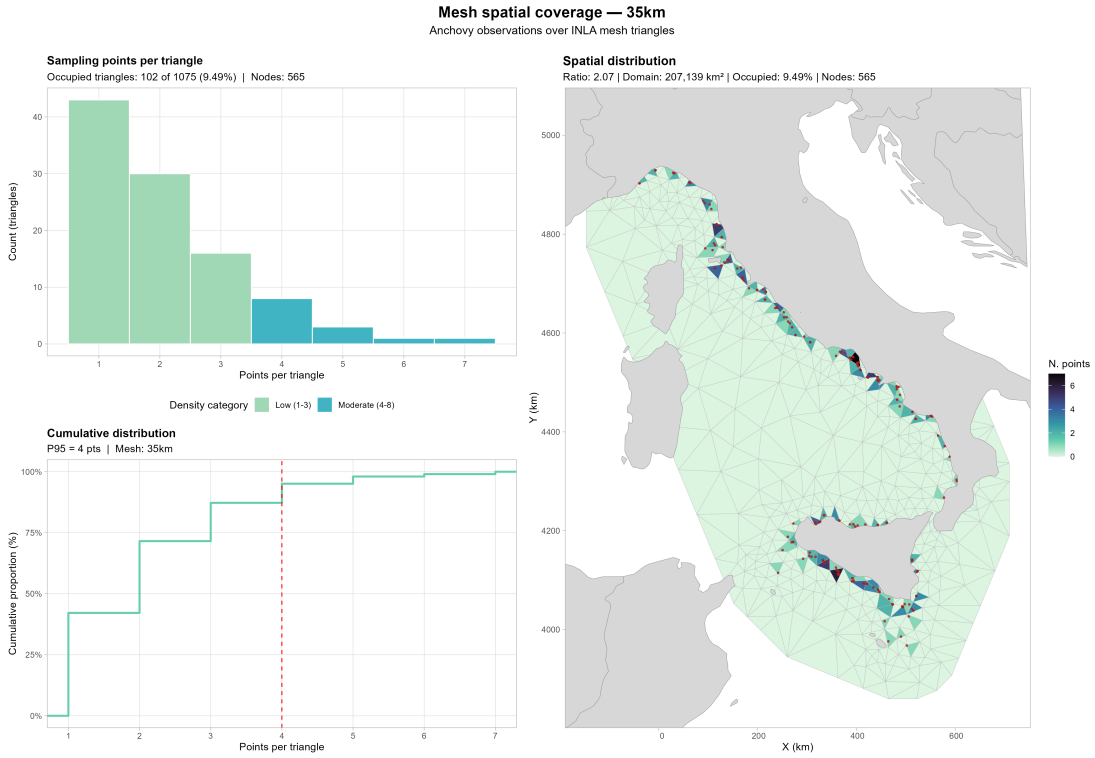
> Note
>
> This is a simple toy example using anchovy presences (2014–2020) in the Mediterranean Sea. The meshes and grids shown here are only used to illustrate how the functions work for both mesh-based and grid-based cases.

4.1 ▲ Mesh coverage

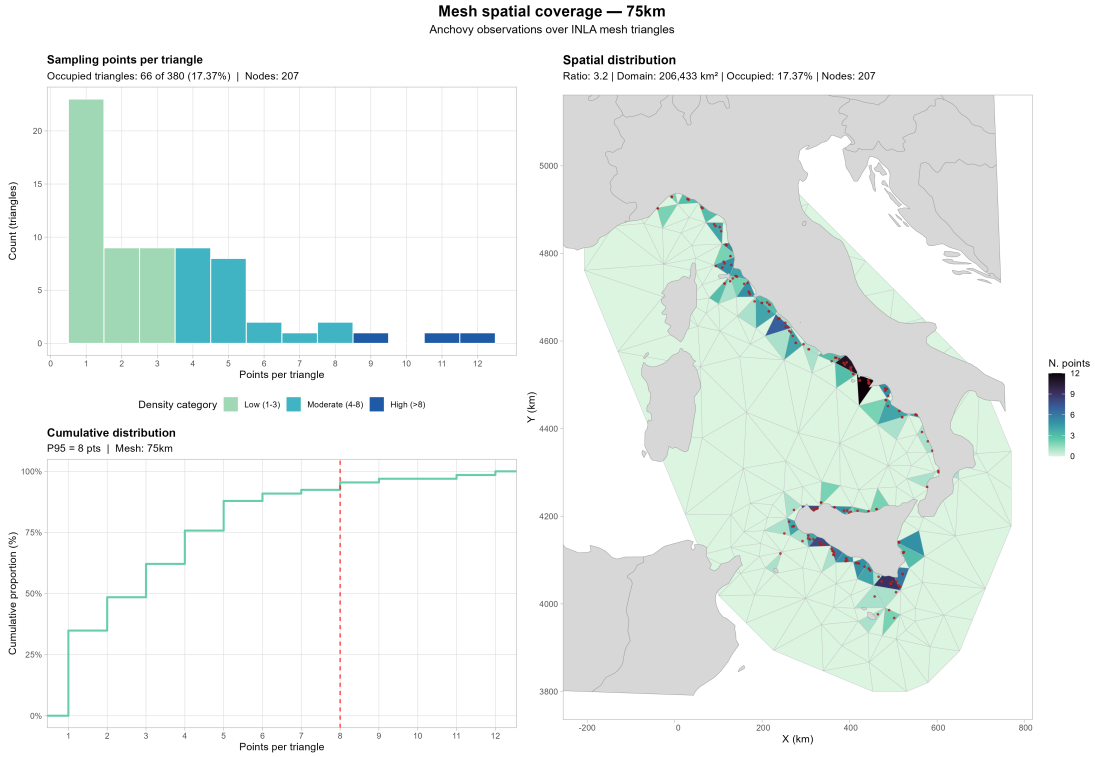
Script: `mesh_coverage.R` · **Use case:** sdmTMB, R-INLA

The INLA mesh divides the spatial domain into **irregular triangles**. We compare two resolutions:

4.1.a Fine mesh — 35 km



4.1.b Coarse mesh — 75 km

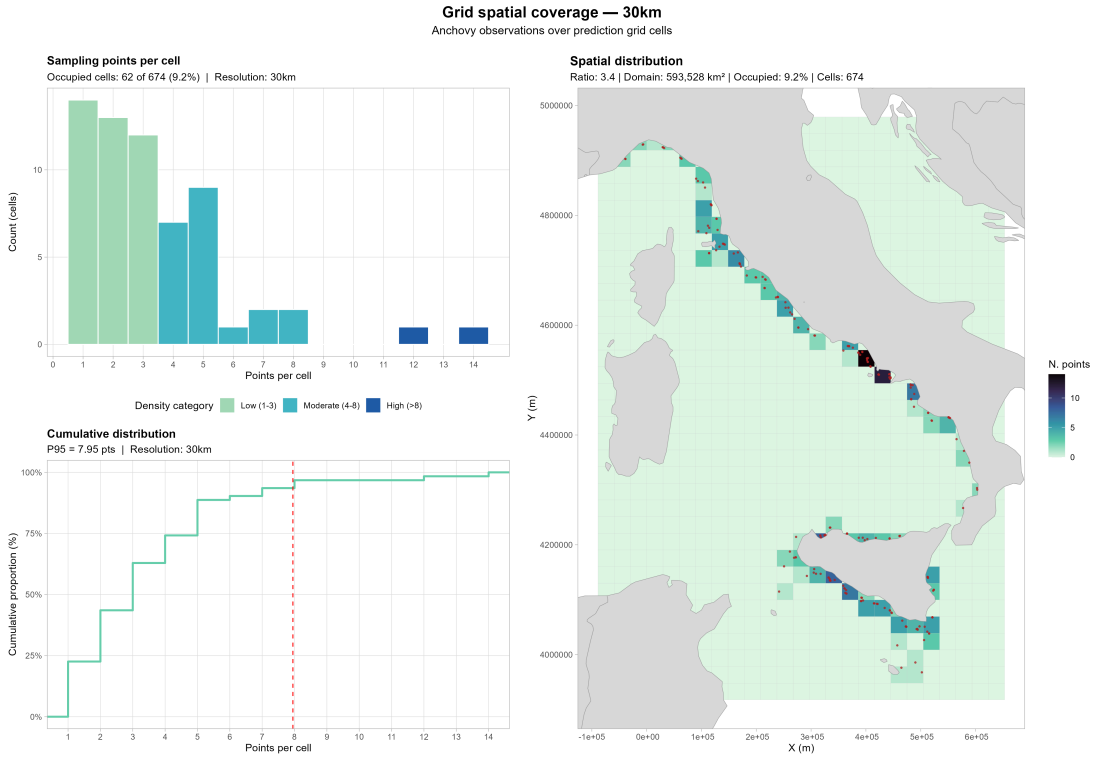


4.2 🔲 Grid coverage

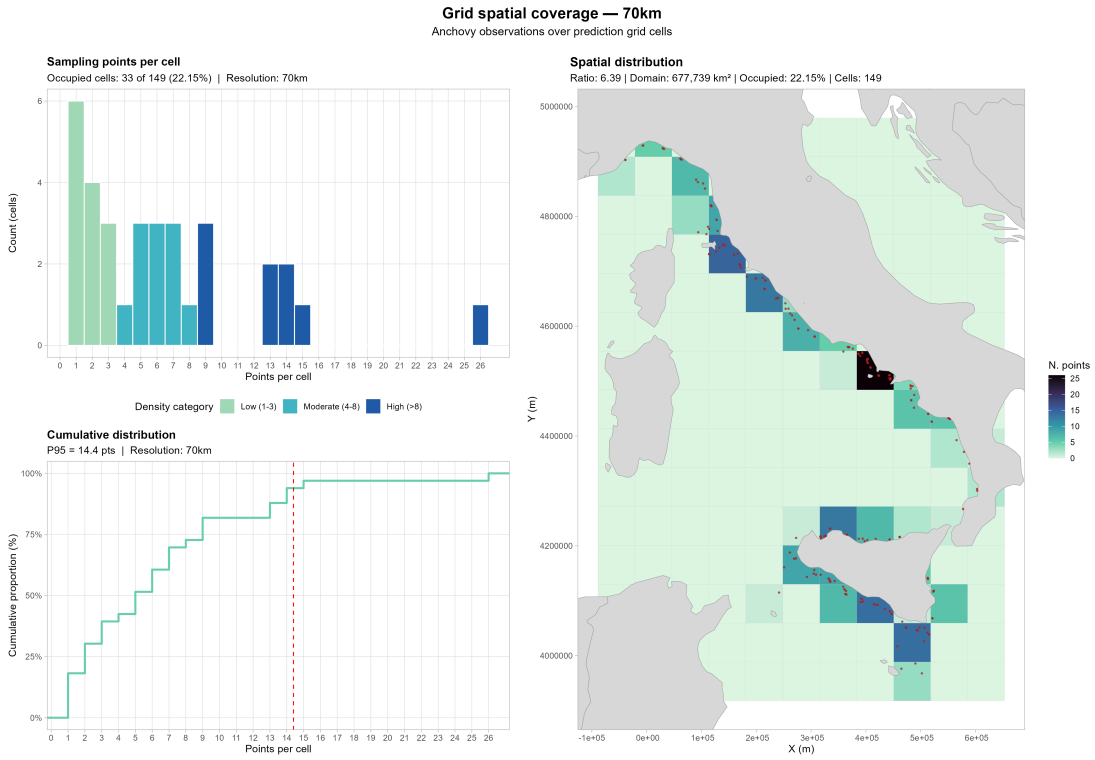
Script: `grid_coverage.R` · **Use case:** GAM, GLM, BART, RF, MaxEnt…

For raster-based SDMs, the prediction domain is a **regular grid**. We compare two resolutions:

4.2.a Medium grid — 30 km



4.2.b Coarse grid — 70 km



4.3 Coverage metrics comparison

4.3.a Mesh coverage

Metric	Fine mesh (35 km)	Coarse mesh (75 km)
% Occupied	9.49%	17.37%
Ratio	2.07	3.2
P95	4	8

4.3.b Grid coverage

Metric	Medium grid (30 km)	Coarse grid (70 km)
% Occupied	9.2%	22.15%
Ratio	3.4	6.39
P95	7.95	14.4